RSS-Based Access Point Localization Experiment

1 Experimental Setup

This experiment evaluates RSS-based localization for a single Wi-Fi access point inside the Faculty of Engineering at Ferdowsi University of Mashhad. The target access point was located in a corridor area of the building. Its reference coordinate, measured independently, was:

`36 deg 18'46.1"N, 59 deg 31'34.9"E`

which corresponds to:

`latitude = 36.3128055556,  longitude = 59.5263611111.`

The experiment uses the pivot RSS table as the only RSS data source:

`data/pivot_signal_table.csv`

The floor map was obtained from:

`data/map.json`

The room and corridor outlines were drawn from the `map_space_coordinate` table in the map JSON file.

2 Dataset Description

Each row in the pivot table contains the measurement position and the RSS values observed from multiple Wi-Fi access points. A value of $-100\,\mathrm{dBm}$ was treated as a missing or non-detected RSS observation. For the selected target access point, the number of valid RSS samples was 1069.

The selected access point was:

`BSSID = b28ba99a8d2a`

Table 1: Summary of RSS samples for the selected access point.

Quantity	Value
Valid RSS samples	1069
Maximum RSS	$-29.0\,\mathrm{dBm}$
Mean RSS	$-72.89\,\mathrm{dBm}$
RSS-weighted centroid error from reference AP	$1.37\,\mathrm{m}$

Figure 1 shows the RSS measurements of the selected access point overlaid on the building map. Stronger RSS values are shown in green, while weaker RSS values are shown in red.

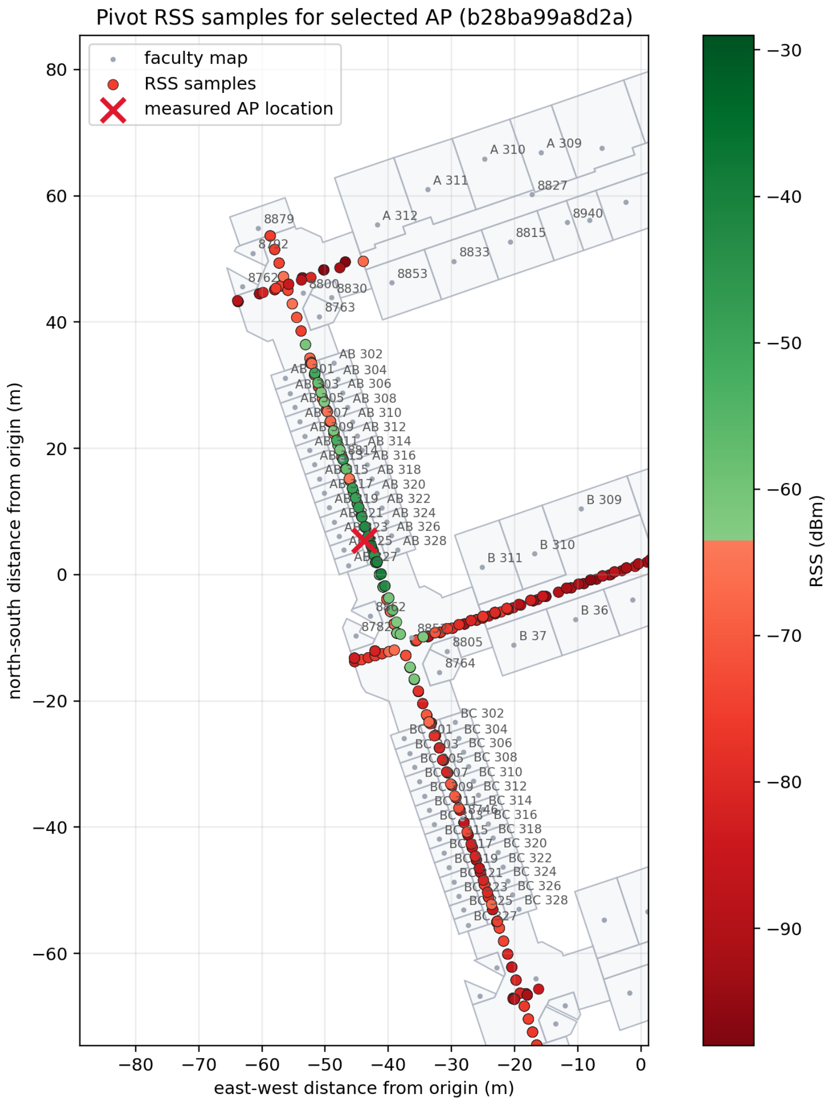

Figure 1: RSS samples of the selected access point overlaid on the faculty map.

3 Access Point Selection

To identify the access point corresponding to the target corridor location, all RSS columns in the pivot table were examined. For each access point, only valid RSS observations, i.e., samples with RSS greater than $-100\,\mathrm{dBm}$, were used. An RSS-weighted spatial centroid was then computed for each candidate access point.

For a sample with RSS value r_i, the weight was defined as:

$$w_i = 10^{(r_i - r_{\max})/10}, \tag{1}$$

where $r_{\max}$ is the strongest RSS observed for that access point. The weighted centroid was computed as:

$$x_c = \frac{\sum_i w_i x_i}{\sum_i w_i}, \qquad y_c = \frac{\sum_i w_i y_i}{\sum_i w_i}. \tag{2}$$

The selected access point was the candidate whose weighted centroid was closest to the reference AP coordinate. This criterion selected the access point `b28ba99a8d2a`.

4 RSS Fitting Model

After selecting the target access point, its location was estimated by fitting a log-distance path-loss model to the RSS measurements. The model is:

$$RSS(d) = RSS_{1m} - 10n \log_{10}(d), \tag{3}$$

where RSS_{1m} is the estimated RSS at one meter, n is the path-loss exponent, and d is the Euclidean distance between a measurement location and the estimated access point position.

The fitted parameter vector was:

$$\theta = (x_{AP}, y_{AP}, RSS_{1m}, n). \tag{4}$$

The optimization minimized the residual between measured RSS and model-predicted RSS. A robust `soft_l1` loss was used to reduce the effect of outliers, which are common in indoor RSS measurements due to multipath propagation, occlusion, and device orientation changes.

5 Localization Results

Table 2 summarizes the localization and model-fitting results.

Table 2: Estimated AP location and fitted RSS model parameters.

Quantity	Value
Estimated latitude	36.3128106191
Estimated longitude	59.5263414073
Distance from reference AP	1.86 m
Estimated RSS_{1m}	−28.36 dBm
Path-loss exponent n	3.22
RSS RMSE	8.53 dB
RSS MAE	6.91 dB

Figure 2 shows the reference AP coordinate and the estimated AP position on the building map. The red cross indicates the reference coordinate, and the black star indicates the estimated AP location.

The estimated position was approximately 1.86 m away from the reference coordinate. Considering the high variability of indoor RSS measurements, this result indicates that the fitted log-distance model can provide a practical approximation of the access point location when sufficient RSS samples are available around the target area.

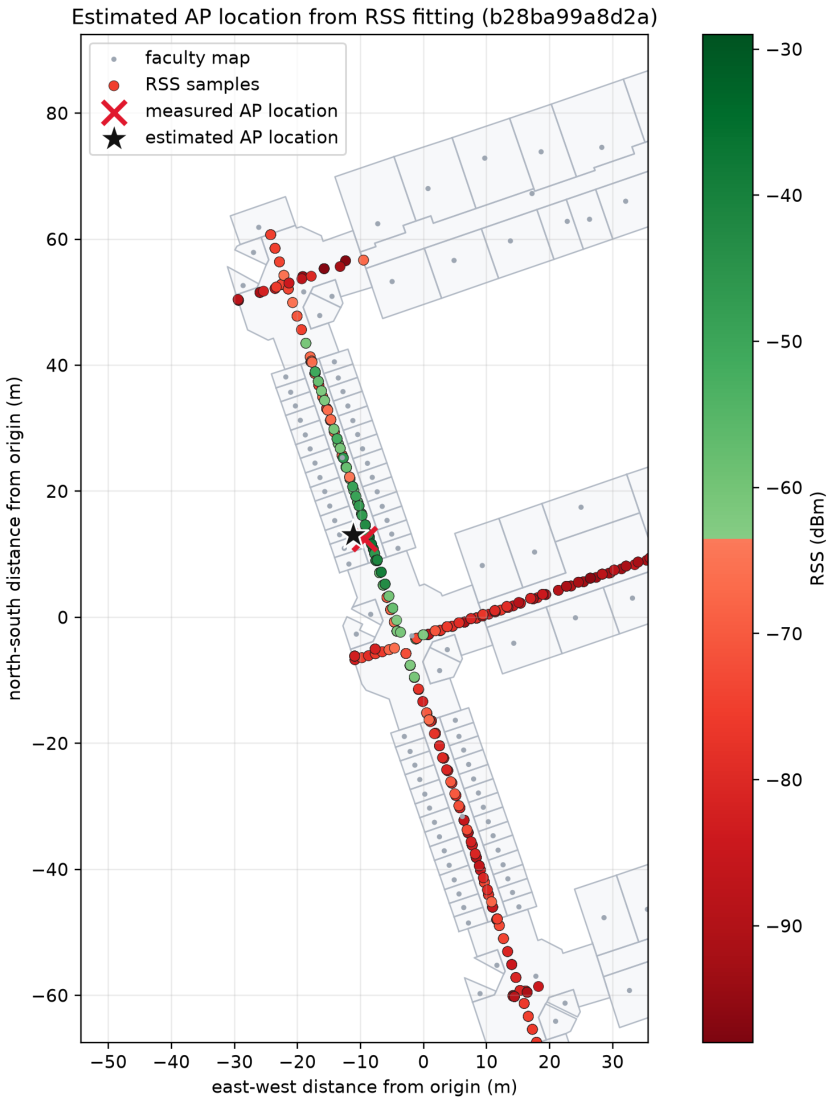

Figure 2: Reference and estimated access point locations on the faculty map.